SafeLink Beacon

Instant Protection. Immediate Response.

One-button emergency safety for people who cannot afford delay.

SMASH National Presentation

Presented by: Joshua Parker, Jordyn Marks, Joshua Young, & Seth Barnes

SAFELINK
Beacon

In case of emergency, simplicity is safety.

No matter the situation, people need quick and efficent access to help.

20 People

Every minute in the U.S. are physically abused by an intimate partner.

29,568

Missing-child reports were made to NCMEC in 2024.

1 in 10

Older adults experience abuse from family each year.

BRAINSTORMING

Stationary Alarm System

Pros:

- Easy Activation
- Easy to locate

Cons:

- No individual safety
- Only works locally

Community Safety Network

Pros:

- Wide safety range
- Quick reporting system

Cons:

- Users don’t keep records
- Doesn’t account for specifics

OUR SOLUTION

SafeLink Beacon:

- Signals help
- Records evidence
- Alerts trusted contacts in seconds

Built for moments when using a phone is too slow, too difficult, or too risky.

One press

No unlocking, searching, or calling.

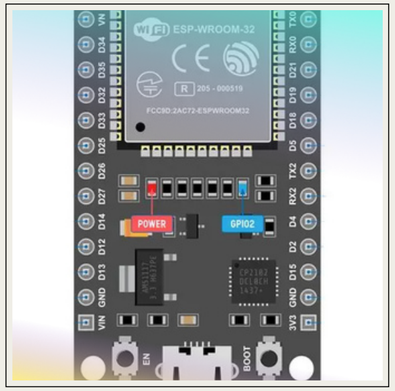
Fast response

Triggers the chosen safety action immediately.

App-connected

Sends alerts, location, and evidence automatically.

PARTS BREAKDOWN



ESP-32

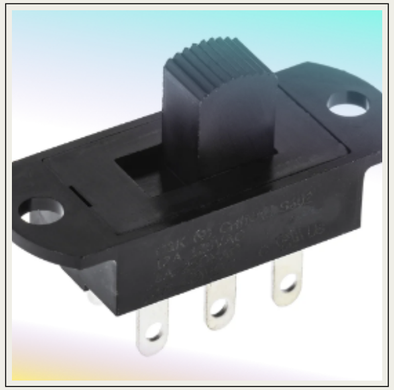
Controls the device and enables smart response logic.

*32-bit microcontroller by Espressif



Colored Pushbuttons

Let the user choose the right safety response.



SlideSwitch

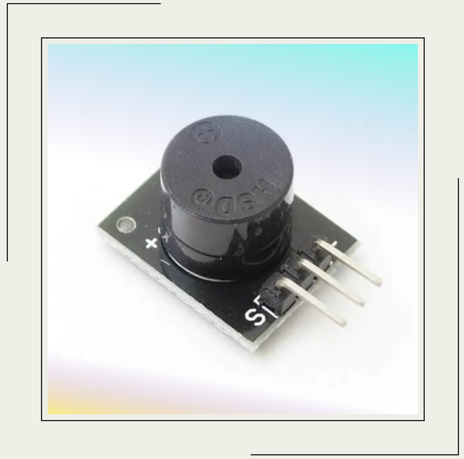
Prevents accidental activation and locks in mode.

PARTS BREAKDOWN



RGB LED

Visual mode feedback for mode activation.

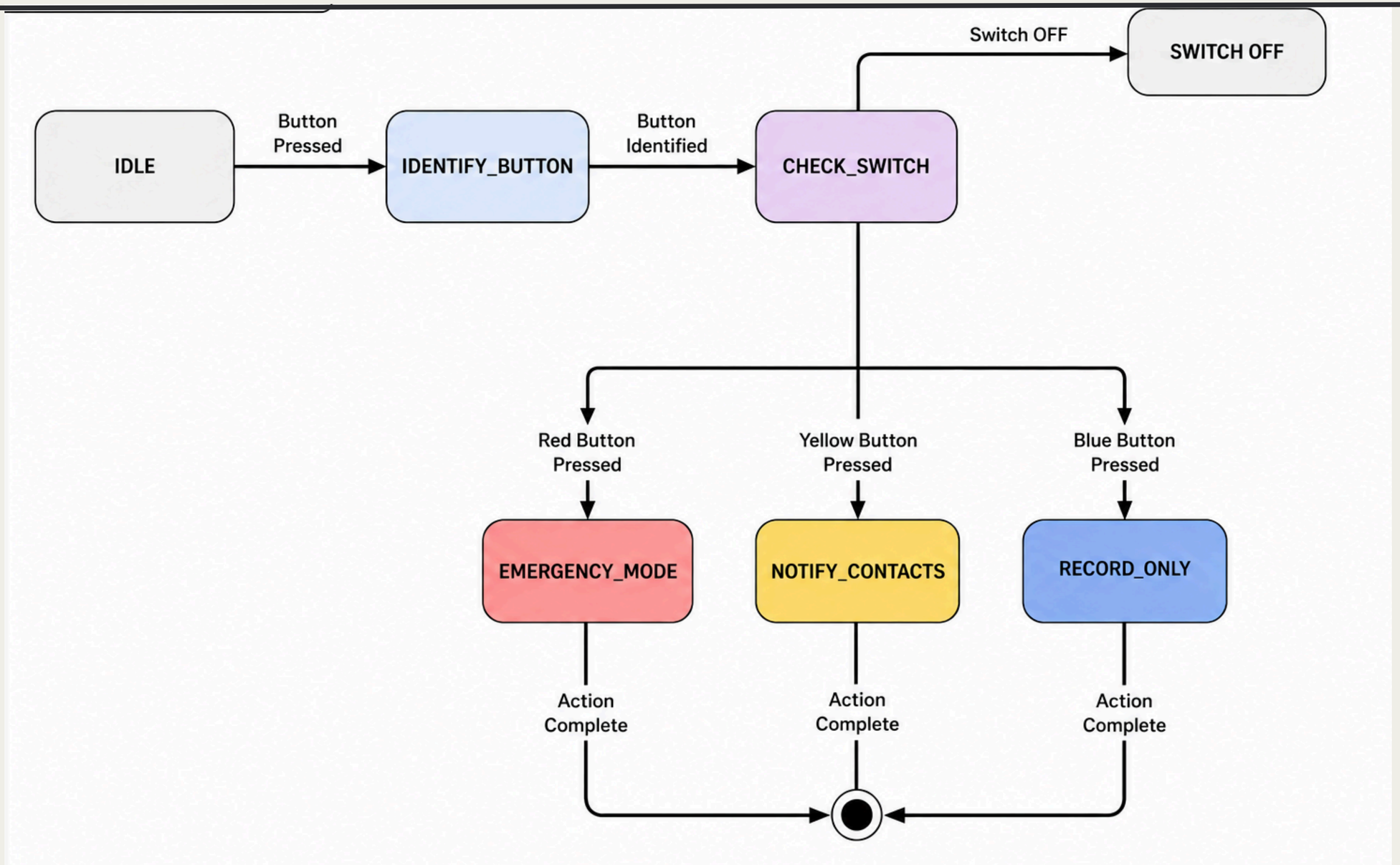


Buzzer

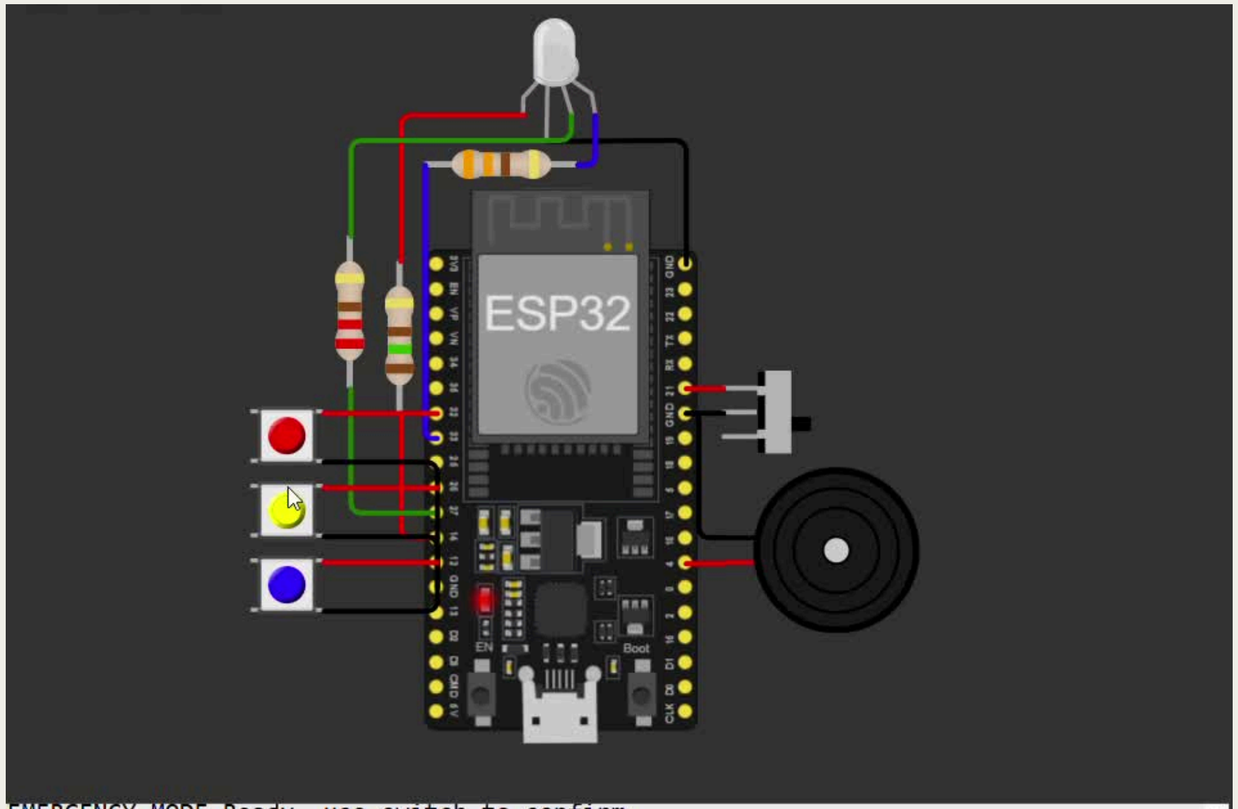
Verbal and physical feedback for activation.

Low-cost solution to make safety affordable.

SYSTEM MECHANICS FINITE STATE MACHINE (FSM)



BEACON DEMO(WOWKI)




```
EMERGENCY MODE Ready, use switch to confirm.
EMERGENCY MODE Ready, use switch to confirm.
EMERGENCY MODE Ready, use switch to confirm.
EMERGENCY MODE Ready, use switch to confirm.
EMERGENCY MODE Ready, use switch to confirm.
EMERGENCY MODE Ready, use switch to confirm.
EMERGENCY MODE Ready, use switch to confirm.
```

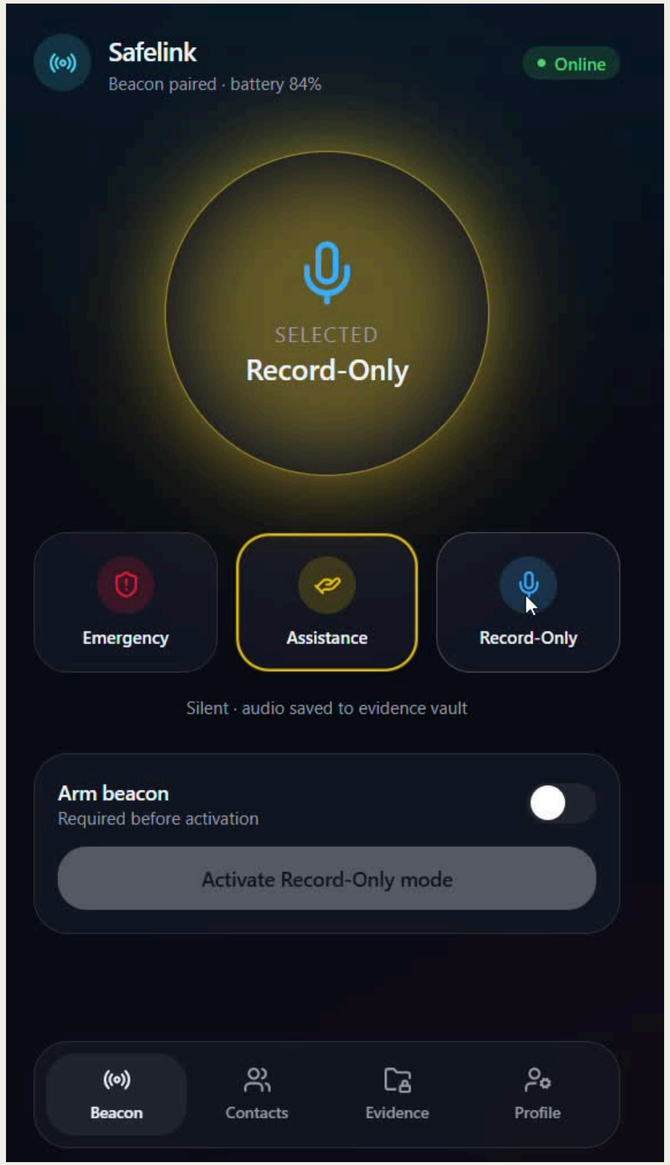


SAFELINK COMPANION APP

SafeLink App:

- Stores & shares Evidence Recordings
- Allows for adding contacts
- Customizes buttons and functionality of Beacon

The device keeps you safe in the moment, and the app manages everything after.



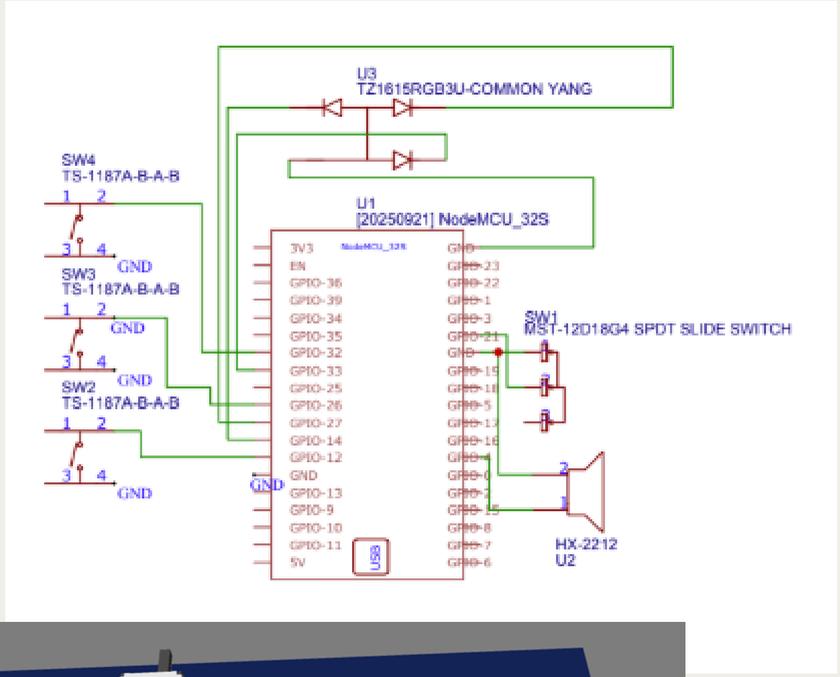
PRODUCT PLANS

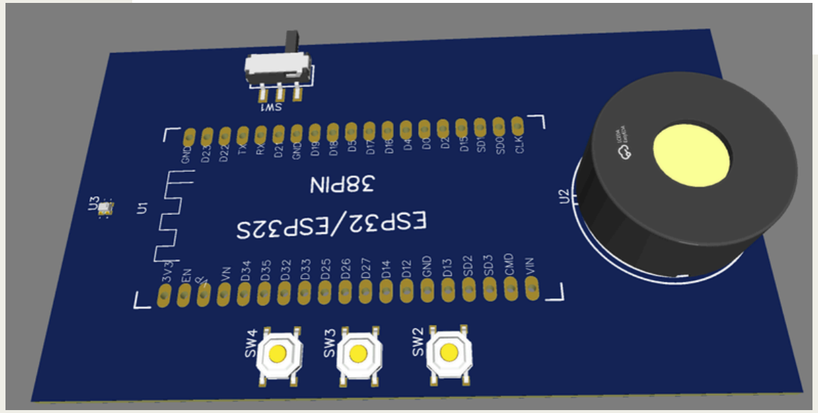
In Development

- Fail-Safe Activation upon breaking
- SIM card usage
- Printed Circuit Board (PCB) Design

Future Developments

- App real-time location
- Health Tracking
- More response options





Thank you!

BUILDING SAFER COMMUNITIES THROUGH TECHNOLOGY- TEAM SAFELINK

SafeLink ESP32

SafeLink Companion

SAFELINK
Beacon